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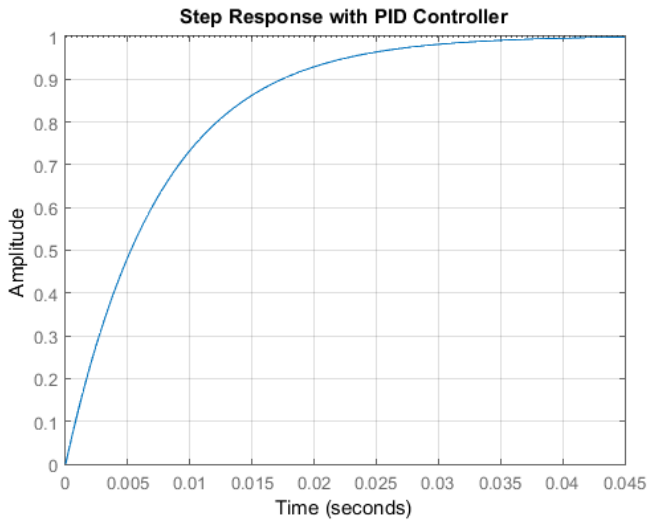
Hasil Project Kontrol PID

Program transfer fuction :

```
clear
clc
% Parameter Motor
R = 0.4;
L = 2.7;
J = 0.0004;
B = 0.0022;
Ke = 0.015;
Kt = 0.05;

% Numerator dan Denominator
num = Ke;
den = [L*J (L*B + R*J) (R*B + Ke*Kt)];

% Transfer Function (Open-loop)
Gs = tf(num, den);

% PID parameters
kp = 53.7208;
Ti = 0.73245;
Td = 0.17711;
ki = kp / Ti;
kd = kp * Td;

% Buat controller PID
PID = pid(kp, ki, kd);

% Closed-loop transfer function (Negative feedback, unity)
T = feedback(PID * Gs, 1);
```

```
% Step response
step(T)
title('Step Response with PID Controller')
grid on

% Hitung karakteristik step response
info = stepinfo(T)

% Tampilkan hasil karakteristik
fprintf('\nKarakteristik Step Response:\n');
fprintf('Rise Time     : %.4f s\n', info.RiseTime);
fprintf('Peak Time     : %.4f s\n', info.PeakTime);
fprintf('Overshoot     : %.2f %%\n', info.Overshoot);
fprintf('Settling Time : %.4f s\n', info.SettlingTime);
```

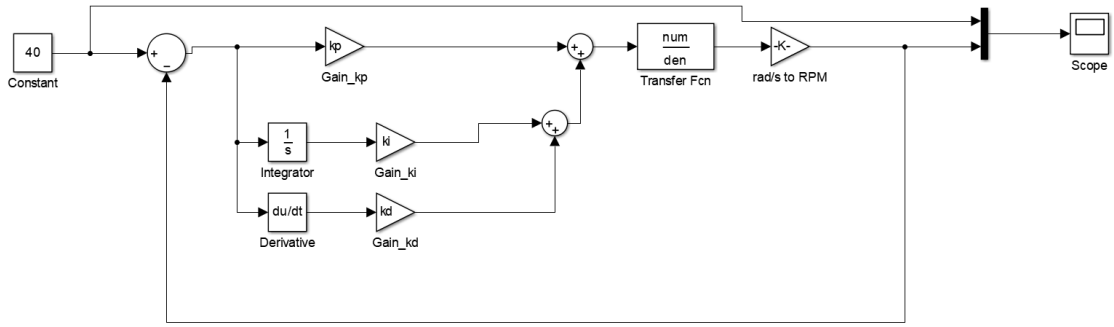
Hasil Run :



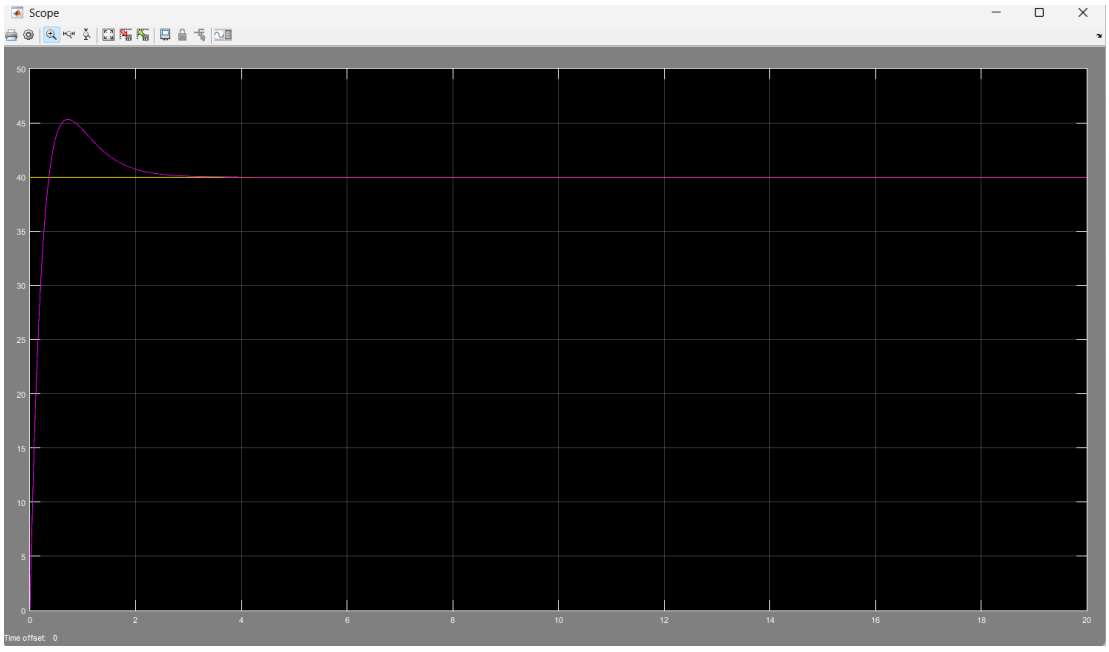
info = RiseTime: 0.0166 SettlingTime: 0.0294 SettlingMin: 0.9002 SettlingMax: 1.0010	Karakteristik Step Response: Rise Time : 0.0166 s Peak Time : 0.0554 s Overshoot : 0.10 % Settling Time : 0.0294 s

Overshoot: 0.1002 Undershoot: 0 Peak: 1.0010 PeakTime: 0.0554	

Simulink :



Hasil Simulink :



Program ARDUINO IDE :

```
#include <WiFi.h>
#include <PubSubClient.h>
// ============================
// KONFIGURASI WIFI & MQTT
// ============================
const char* ssid = "TKA 6Barokah";
const char* password = "65432100";
const char* mqtt_server = "broker.hivemq.com";
const int mqtt_port = 1883;
WiFiClient espClient;
PubSubClient client(espClient);
// ============================
// PIN KONFIGURASI
// ============================
const int pinMotorPWM = 5;
const int pinREn = 21;
const int pinLEn = 22;
const int encoderPinA = 34;
const int encoderPinB = 35;
// ============================
// VARIABEL PID & MOTOR
// ============================
kp = 53.7208;
Ti = 0.73245;
Td = 0.17711;
ki = kp / Ti;
kd = kp * Td;
float error = 0, lastError = 0, integral = 0;
float targetSpeedKmh = 0;
float actualSpeedKmh = 0;
int pidPWM = 0;
int throttleLevel = 0;
String motorDirection = "forward"; // Default arah
```

```
// ============================
// ENCODER
// ============================
volatile long encoderCount = 0;
unsigned long lastSpeedCheck = 0;
void IRAM_ATTR encoderISR() {
  encoderCount++;
}
// ============================
// WIFI DAN MQTT SETUP
// ============================
void setup_wifi() {
  Serial.print("Connecting to WiFi ");
  Serial.print(ssid);
  WiFi.mode(WIFI_STA);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println();
  Serial.println("WiFi connected, IP: " + WiFi.localIP().toString());
}

void reconnect() {
  while (!client.connected()) {
    Serial.print("Connecting to MQTT...");
    String clientId = "ESP32Client-";
    clientId += String(random(0xffff), HEX);
    if (client.connect(clientId.c_str())) {  // Connect tanpa user/password
      Serial.println("connected!");
      client.subscribe("motor/throttle");
      client.subscribe("motor/direction");
    } else {
      Serial.print("failed, rc=");
```

```
      Serial.print(client.state());
      Serial.println(" try again in 5 seconds");
      delay(5000);
    }
  }
}
// ============================
// CALLBACK MQTT
// ============================
void callback(char* topic, byte* payload, unsigned int length) {
  String msg;
  for (unsigned int i = 0; i < length; i++) {
    msg += (char)payload[i];
  }

  Serial.print("[MQTT] Topic: ");
  Serial.print(topic);
  Serial.print(" | Message: ");
  Serial.println(msg);

  if (String(topic) == "motor/throttle") {
    int throttle = msg.toInt();
    throttleLevel = constrain(throttle, 0, 3);
    switch (throttleLevel) {
      case 1: targetSpeedKmh = 5; break;
      case 2: targetSpeedKmh = 10; break;
      case 3: targetSpeedKmh = 15; break;
    }
    Serial.println("[MQTT] Throttle level set: " + String(throttleLevel));
  }

  if (String(topic) == "motor/direction") {
    msg.toLowerCase();
    if (msg == "forward" || msg == "reverse" || msg == "neutral") {
      motorDirection = msg;
```

```
      Serial.println("[MQTT] Direction set: " + motorDirection);
    }
  }
}

// ============================
// SETUP
// ============================
void setup() {
  Serial.begin(115200);

  pinMode(pinREn, OUTPUT);
  pinMode(pinLEn, OUTPUT);

  ledcSetup(0, 5000, 8); // 5kHz, 8-bit PWM
  ledcAttachPin(pinMotorPWM, 0);

  pinMode(encoderPinA, INPUT_PULLUP);
  pinMode(encoderPinB, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(encoderPinA), encoderISR, RISING);

  setup_wifi();
  client.setServer(mqtt_server, mqtt_port);
  client.setCallback(callback);
}

// ============================
// MENGHITUNG KECEPATAN
// ============================
void updateSpeed() {
  noInterrupts();
  long count = encoderCount;
  encoderCount = 0;
  interrupts();
```

```
  float rpm = (count / 20.0) * 60.0; // 20 pulses per rotation
  float wheelCircumference = 3.14 * 0.1; // Diameter roda 10 cm
  actualSpeedKmh = (rpm * wheelCircumference * 60.0) / 1000.0;

  Serial.println("[ESP32] Actual Speed: " + String(actualSpeedKmh, 2) + " km/h");
  client.publish("motor/actual_speed", String(actualSpeedKmh, 2).c_str());
}

// ============================
// KONTROL ARAH
// ============================
void controlDirection() {
  if (motorDirection == "forward") {
    digitalWrite(pinREn, HIGH);
    digitalWrite(pinLEn, LOW);
  } else if (motorDirection == "reverse") {
    digitalWrite(pinREn, LOW);
    digitalWrite(pinLEn, HIGH);
  } else { // neutral
    digitalWrite(pinREn, LOW);
    digitalWrite(pinLEn, LOW);
  }
}

// ============================
// PID CONTROL
// ============================
void PIDControl() {
  if (motorDirection == "neutral" || targetSpeedKmh == 0) {
    ledcWrite(0, 0);
    pidPWM = 0;
    integral = 0;
    lastError = 0;
    return;
  }
```

```
  error = targetSpeedKmh - actualSpeedKmh;
  integral += error;
  float derivative = error - lastError;
  lastError = error;

  float output = Kp * error + Ki * integral + Kd * derivative;
  pidPWM = constrain((int)output, 0, 255);
  ledcWrite(0, pidPWM);

  Serial.println("[ESP32] PID PWM: " + String(pidPWM));
  client.publish("motor/pid_output", String(pidPWM).c_str());
}

// ============================
// LOOP UTAMA
// ============================
void loop() {
  if (!client.connected()) {
    reconnect();
  }
  client.loop();

  controlDirection();

  unsigned long now = millis();
  if (now - lastSpeedCheck >= 1000) {
    lastSpeedCheck = now;
    updateSpeed();
    PIDControl();
  }
}
```